Nicholas Chen

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EDUCATION

University of California at Berkeley — **Berkeley, California**
B.S. in Electrical Engineering and Computer Science — *Expected Graduation, May 2028*

- **GPA:** 3.94/4.0 (Junior Standing)
- **Related Coursework:** Linear Algebra and Differential Equations (MATH54), The Structure and Interpretation of Computer Programs (CS61A), Data Structures (CS61B), Discrete Mathematics and Probability Theory (CS70).

EXPERIENCE

MIT Pentelute Lab — **Boston, Massachusetts**
Research Intern — *Feb 2023 – Apr 2024*

- Developed a stochastic generative modeling + optimization stack for peptide candidates using diffusion models (sampling) and deep reinforcement learning (policy/value optimization) in PyTorch with vectorized preprocessing in NumPy/Pandas.
- Designed the RL reward as a multi-objective utility: maximize predicted affinity and developability scores while penalizing constraint violations (aggregation/toxicity heuristics), turning peptide design into a constrained optimization loop.
- Improved reproducibility by tracking configs, data lineage, training (loss/grad norms), RL dynamics (reward curves/Q estimates), and generation KPIs (top-k scores, diversity/uniqueness, sequence-level stats) across sensitivity analyses.
- Published a research paper with Dr. Vladimir Akhmetov on AI-assisted Alzheimer's drug design (DOI: 10.36838/v7i4.5).

Oxford University — **Oxford, United Kingdom**
Academic Researcher — *Dec 2023 – Jan 2024*

- Ranked 1/45 in a research team modeling biochemical systems through stochastic processes and algorithmic complexity.
- Implemented Monte Carlo simulation and time-series / nonlinear dynamical systems models in Python, C++, and MATLAB, including parameter sweeps and stability/sensitivity analysis of molecular interaction dynamics.
- Built high-dimensional statistical learning pipelines in NumPy/Pandas/scikit-learn, applying PCA (numerical linear algebra), k-means (unsupervised clustering), and Random Forest for candidate scoring and uncertainty/risk-style assessment.

UC Berkeley Operations & Behavioral Analytics Lab — **Berkeley, California**
AI/Game Development Engineer — *Sep 2025 – Present*

- Modeled sequential decision-making under uncertainty in delayed-reward environments using Deep Q-Networks (DQN); improved policy learning via reward shaping, experience replay, target networks, and hyperparameter sweeps.
- Built an experiment-grade simulation platform in SvelteKit with Firebase (auth, realtime streaming) to collect event-level trajectories and state/action/reward logs for policy evaluation and behavioral analysis (https://bit.ly/45kvN6r).
- Quantified behavioral effects in human–agent interaction data using controlled comparisons and uncertainty estimates (confidence intervals / bootstrap) over key outcome metrics; presented with Prof. Park Sinchaisri at Stanford and CMU..

Shenwan Hongyuan — **Hong Kong, SAR**
Data Engineering Intern — *Jan 2025 – Feb 2025*

- Owned market-data QA + reconciliation for trading pipelines; implemented schema/null/outlier and symbol/timestamp alignment checks with researchers; optimized Pandas/NumPy/Spark transforms for 20% faster runtime.
- Automated and hardened ETL for time-series market feeds (Python/SQL/Airflow): idempotent jobs, backfill-safe runs, and dependency-managed DAGs; improved end-to-end ingestion/processing throughput by 10%.
- Tuned PostgreSQL performance for production queries (partition-aware access patterns, query-plan optimization), cutting query latency by 30%; added Redis caching for hot datasets and containerized services with Docker for deployment.

PUBLICATIONS & PROJECTS

CivicGrid - Voice-to-Structured Data Pipeline *(https://github.com/nnicholas-c/CivicGrid)* — *CalHacks 2025, Featured at YC Afterparty*

- Built a full-stack voice triage pipeline (React/TypeScript + Flask) using WebSockets for real-time call transcription (Deepgram) and structured work-order creation, deployed end-to-end across GitHub Pages + Railway + Firebase.
- Productionized an LLM/vision normalization layer (Grok API) that outputs schema-valid JSON with priority/severity scoring, geo/address extraction, and multi-class issue categorization, enabling fast query/filtering of civic incident records.

Stock Market Prediction Web App (*https://github.com/nnicholas-c/Python-stock-predictor)* — *May 2025*

- Implemented a modular data, feature engineering, model training, and inference pipeline in Python/Flask + scikit-learn; served predictions via a REST API to a React frontend with live Alpha Vantage data; achieved MAE 0.16 and R^2 0.98.

HONORS & AWARDS

- 2022-2024 New Zealand Chemistry Olympiad 2x Gold Medal, 1x Silver Medal; National Exam Full Score Achiever;
- 2024 International Chemistry Olympiad Qualifier; New Zealand Chemistry Olympiad National Team Member;
- 2024 New Zealand Mathematical Olympiad Camp Member; New Zealand Senior Maths Competition 1st Place;
- 2024 British Biology Olympiad Gold Medal; British Physics Olympiad Gold Medal; UK Chemistry Olympiad Gold Medal;
- 2024 New Zealand Young Physicists' Tournament 2nd Place; 2023 New Zealand Young Physicists' Tournament 3rd Place;
- 2023 New Zealand Physics and Maths Competition 7th place; International Young Physicists' Tournament Qualifier;

SKILLS

Programming language: Python, JavaScript, Java, C++, SQL, C#
Web Dev: Flask, MySQL, HTML/CSS
Machine learning: Pytorch, Scikit-Learn, Numpy, Pandas
Others: Git, Bash, Hugging Face